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Program: Software Engineering

Operating Systems

Laboratory Work No. 1

The OS as a Resource Manager

Track A: The Real OS, from the Outside (Linux)

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1. Introduction

The purpose of this laboratory work is to observe how an operating system manages the main resources of a computer system.

The experiment was performed using Ubuntu running under Windows Subsystem for Linux 2 (WSL2). During the laboratory work, four major areas of operating system resource management were investigated: files and directories, processes, memory, and devices/storage.

2. System Information

The following commands were used to identify the current user, operating system kernel, and system uptime:

```
whoami
uname -a
uptime
```

The system reported the following information:

```
Linux DESKTOP-10DB5P6 5.15.167.4-microsoft-standard-WSL2
#1 SMP Tue Nov 5 00:21:55 UTC 2024
x86_64 x86_64 x86_64 GNU/Linux

17:26:07 up 2 min, 1 user,
load average: 0.01, 0.00, 0.00
```

The Linux environment was running using the WSL2 Linux kernel.

3. Files and Directories

In the first part of the laboratory work, the Linux file system was explored and basic file-management operations were performed.

Commands such as the following were used:

```
pwd
ls -la /
ls -la ~
cd /etc
ls | head
```

A demonstration directory and several files were then created:

```
mkdir demo
cd demo
```

```
echo "hello operating systems" > note.txt
cp note.txt copy.txt
mv copy.txt renamed.txt
ls -l
rm renamed.txt
```

3.1. Observations

1. The files created during the experiment were owned by user **sofia** and belonged to group **sofia**.
2. After executing

```
chmod 600 note.txt
```

the file permissions became:

```
-rw-----
```

The first character, **-**, indicates that **note.txt** is a regular file.

The next three characters, **rw-**, indicate that the owner can read and write the file but cannot execute it.

The following **--** indicates that members of the group have no permissions, while the final **--** indicates that other users also have no permissions.

3. Two directories directly under the root directory **/** are:
 - **/etc** – contains system-wide configuration files.
 - **/home** – contains users' home directories.

3.2. Interesting Command Output

The command **ls -l** displayed the following result:

```
total 8
-rw-r--r-- 1 sofia sofia 24 Sep 28 17:28 note.txt
-rw-r--r-- 1 sofia sofia 24 Sep 28 17:28 renamed.txt
```

This output demonstrates that Linux stores information about ownership, permissions, size, and modification time for every file.

4. Processes

The second part focused on process management.

The currently running processes were inspected using:

```
ps aux | head
ps aux | wc -l
top
```

A new process was created in the background:

```
sleep 300 &
```

It was inspected using:

```
jobs
ps -ef | grep sleep
```

The process was then terminated using the `kill` command.

The Linux pseudo-filesystem `/proc` was also used to inspect information maintained by the kernel about the process.

4.1. Observations

1. Process number 1 has PID 1. On the tested system it was `/sbin/init`.

This is the `init` process responsible for starting and managing system services.

2. Approximately **29 processes** were running while the virtual Linux environment was mostly idle.
3. For the test `sleep` process, the file `/proc/665/status` contained:

```
State: S (sleeping)
```

This means that the process existed but was waiting rather than actively using CPU time.

4.2. Interesting Command Output

The process status contained the following information:

```
Name:   sleep
State:  S (sleeping)
Pid:    665
PPid:   287
VmPeak: 3128 kB
```

```
VmSize:      3128 kB
```

This shows that the operating system maintains detailed information about every process and exposes part of this information through the `/proc` filesystem.

5. Memory Management

The third part of the laboratory work investigated how Linux manages physical memory and swap space.

The following commands were executed:

```
free -h
cat /proc/meminfo | head -6
```

The resident memory used by a simple `sleep` process was inspected with:

```
grep VmRSS /proc/$PID/status
```

5.1. Observations

1. The system had approximately **13 GiB of total RAM**, with about **12 GiB free** at the time of observation.
2. Swap is storage space that can be used by the operating system as an extension of physical memory when RAM is under pressure.

The tested system had **4.0 GiB of swap** configured, and none of it was being used during the experiment.

3. The `sleep` process used approximately **1080 kB** of resident memory according to its `VmRSS` value.

This amount is relatively small, which is expected because the `sleep` process performs almost no computation and requires very little memory.

5.2. Interesting Command Output

	total	used	free	shared	buff/cache	
available						
Mem:	13Gi	711Mi	12Gi	3.3Mi	524Mi	12
Gi						
Swap:	4.0Gi	0B	4.0Gi			

The output demonstrates how the operating system divides physical memory between currently used memory, free memory, and cached data.

6. Devices and Storage

The final experimental part focused on storage devices, mounted filesystems, and device interfaces.

The following commands were used:

```
df -h
lsblk
du -sh ~/os-lab1
ls -l /dev | head
mount | head
```

The device containing the root filesystem was additionally checked using:

```
df -h /
```

6.1. Observations

1. The root filesystem `/` was mounted on:

```
/dev/sdc
```

2. One example of a device interface found inside `/dev` is `/dev/console`.

It represents the system console and provides a file-like interface for terminal input and output.

3. The UNIX principle “everything is a file” means that many operating system resources, including hardware devices, are exposed through file-like interfaces that programs can access using common read and write operations.

6.2. Interesting Command Output

The `lsblk` command showed the block devices visible to Linux:

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
sda	8:0	0	388.4M	1	disk	
sdb	8:16	0	4G	0	disk	[SWAP]
sdc	8:32	0	1T	0	disk	/mnt/wslg/distro
						/

The output shows that `sdb` was used for swap, while `sdc` contained the root filesystem.

7. Conclusion

The operating system manages four major categories of resources: **files, processes, memory, and devices/storage**.

File management was observed using commands such as **ls**; process management was inspected using **ps**; memory usage was examined using **free**; and storage devices were inspected using **lsblk**.

The laboratory work demonstrated how Linux provides structured and controlled access to system resources and acts as an intermediary between running programs and the underlying hardware.